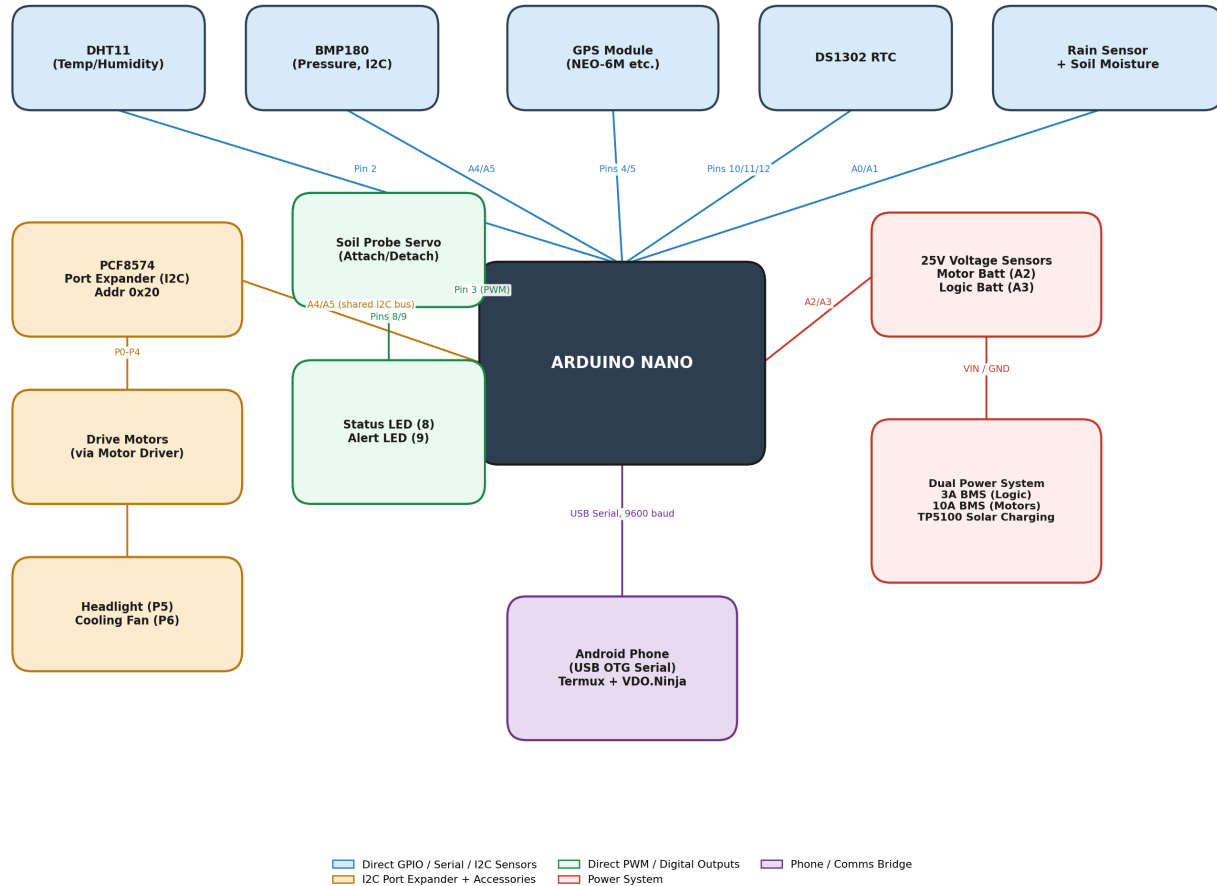


Pragun Smart Rover — Wiring Reference

Full pin mapping and block diagram for the Arduino Nano-based rover controller

Pragun Smart Rover — Wiring Overview

Block diagram — see the pin table for exact wire-by-wire connections



Complete Pin Mapping Table

Component	Arduino Nano Pin	Interface	Notes
DHT11 (Temperature/Humidity)	Pin 2	Digital (1-wire)	DHTTYPE: DHT11
BMP180 (Barometric Pressure)	A4 (SDA), A5 (SCL)	I2C	Shared I2C bus with PCF8574
GPS Module	Pin 4 (RX), Pin 5 (TX)	SoftwareSerial, 9600 baud	Module TX to Nano Pin 4; Module RX to Nano Pin 5
DS1302 RTC	Pin 10 (RST), 11 (DAT), 12 (CLK)	3-wire (ThreeWire)	Keeps timestamp for telemetry
Rain Sensor	A0	Analog	—
Soil Moisture Sensor	A1	Analog	—
Motor Battery Voltage Sensor	A2	Analog (25V module, 5:1 divider)	Formula: $V = (\text{raw}/1023.0) * 25.0$
Logic Battery Voltage Sensor	A3	Analog (25V module, 5:1 divider)	Formula: $V = (\text{raw}/1023.0) * 25.0$
PCF8574 Port Expander	A4 (SDA), A5 (SCL)	I2C, Addr 0x20	Drives motors, headlight, and fan (see table below)
Soil Probe Servo	Pin 3	PWM (direct, not via expander)	Attaches only while moving, detaches after -- prevents idle buzz
Status LED	Pin 8	Digital Output	—
Alert LED	Pin 9	Digital Output	—
Android Phone (Base Station)	USB (Serial)	USB OTG, 9600 baud	Runs Termux bridge and VDO.Ninja camera app

PCF8574 Port Expander — Output Bit Mapping

PCF8574 Pin	Function	Notes
P0	Motor Forward (part of pair)	Combined with P2 for forward
P1	Motor Reverse / Left (part of pair)	Combined with P4 for reverse; with P2 for left
P2	Motor Forward / Left (part of pair)	Combined with P0 for forward; with P1 for left
P3	Unused	Reserved
P4	Motor Reverse / Right (part of pair)	Combined with P1 for reverse; with P0 for right
P5	Headlight	Preserved across motor state changes
P6	Cooling Fan	Preserved across motor state changes; auto-triggers above 32C
P7	Unused	Reserved

Note: the I2C bus (A4/A5) is shared between the BMP180 sensor and the PCF8574 port expander. Both devices sit on the same two wires; make sure pull-up resistors are present on SDA and SCL (many breakout boards include them; a bare PCF8574 chip usually does not). The PCF8574's default address is 0x20 only when its A0/A1/A2 address pins are all tied to GND — if wired differently, update PCF8574_ADDR in the sketch to match.